

Simulating Future Flood Insurance

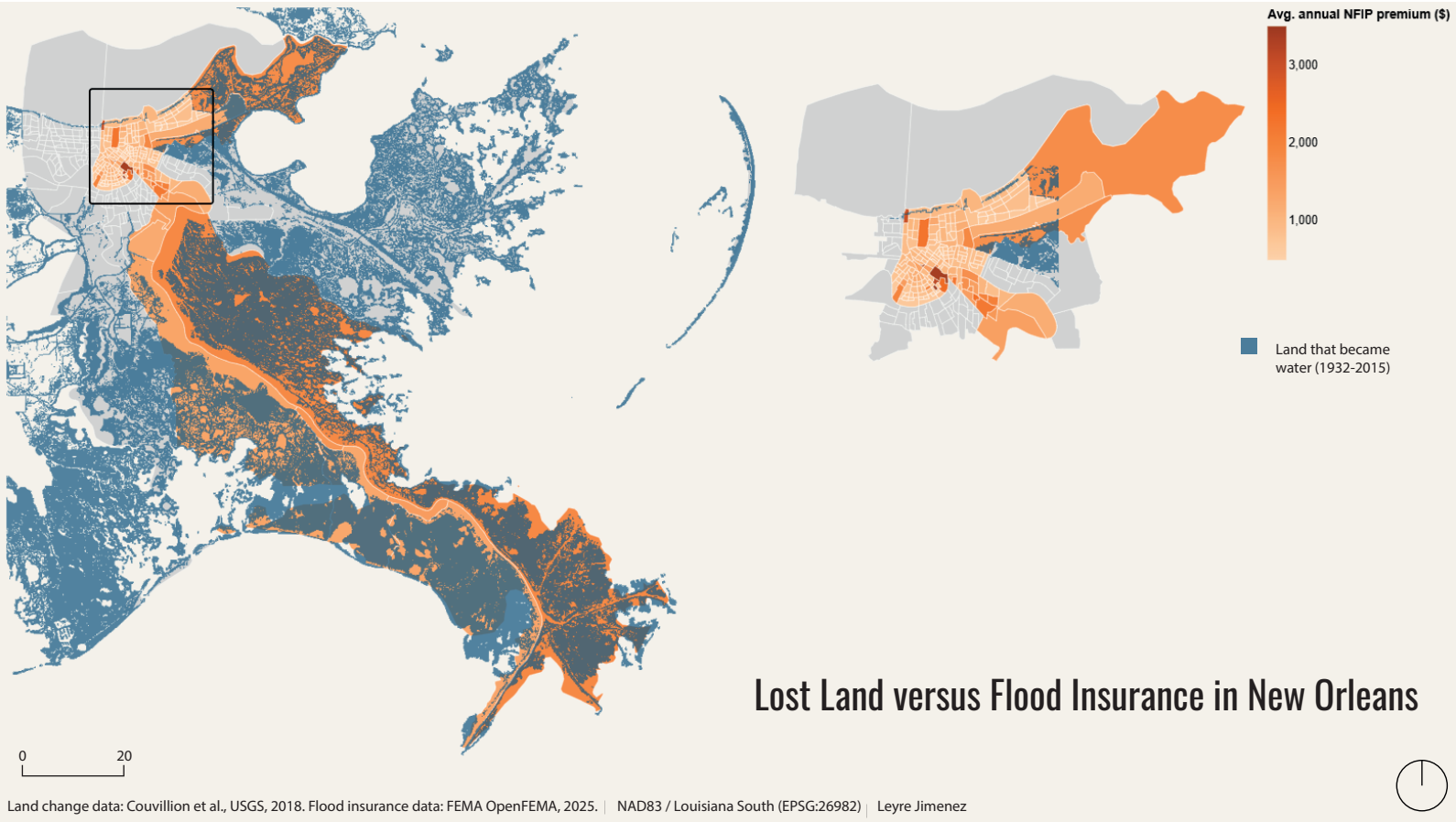
Based on my assignment 2 map below, my simulation would focus on simulating future flood insurance access under coastal land loss in New Orleans.

1. What am I attempting?

I would want to simulate how flood insurance coverage could change between 2025 - 2045 (possibly more) as costal New Orleans continues to lose land. In assignment 2, my map compared land that became water (1932-2015) with FEMA flood insurance premium data from 2025. Instead of thinking that insurance will always rise as there is more land loss and water gets closer, I would want to ask as projected new water gets closer to communities, where might insurance coverage decline, remain or be preserved by coastal protection/restoration.

The model would be based on the diea that rising climate risk is not necesarily correspondant with more inrance coverage. In other areas, with more climate risk, insurers have raised premiums, created non renewal policies or left from the area because it is too risky. The simulation would not claim that NFIP behaves like provite homeowever but it uses insurance withdrawal as a scenario. So the model would ask: if acess to coverage declines in places where risk increases, where would the larges gaps/mismatches be between enviromental exposure and insurance occur?

The simulation would not predict any behavior or the individual household, floor damage or future insurance enrolmment. The scenario model makes assumptions and then asks how these assumptions produce different patterns.



2. The Dataset

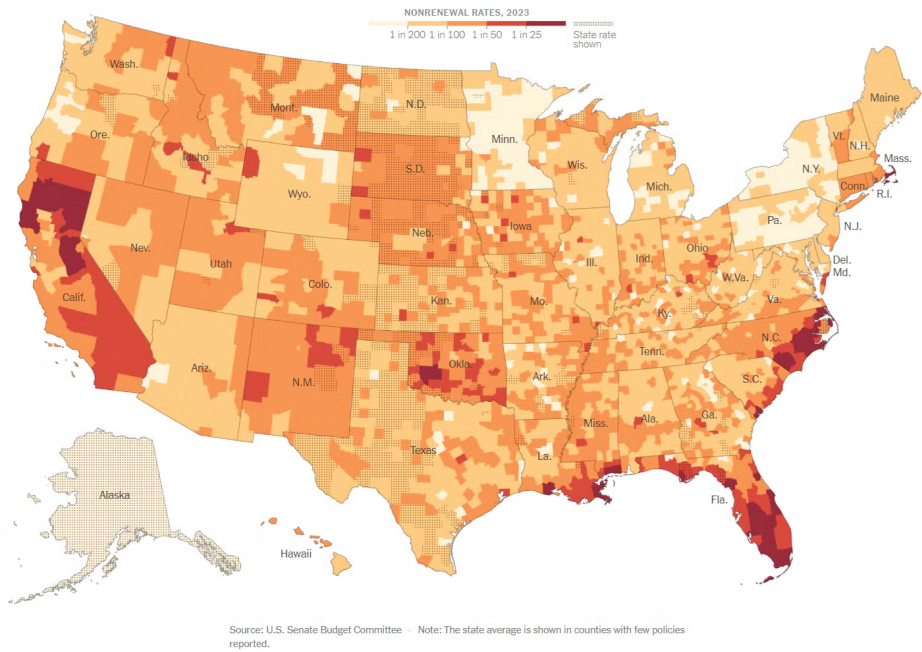
I will use the historic land change data from USGS from 2017. The data spans from 1932 to 2016 and measures land and water changes in coastal Louisiana using satellite imagery and surveys. It records where land was converted to water.

For this simulation, each land-change record will be one [grid cell / polygon / coastal area] from that dataset. The columns used would be: land_id, geometry, start_year, end_year, land_to_water, land_area_lost, historic_loss_rate and parish.

I would also use data from FEMA NFIP policies dataset from 2025. The dataset contains policy transactions from the National Flood Insurance Program. Policy records include dates and generalized geographic fields that can be aggregated to places such as census tracts, ZIP codes, counties, or parishes. Each insurance row will have one census tract or zip code. The columns used would be: geography_id, year, active_NFIP_policy, total_coverage, average_premium, historic land loss, projected land loss, distance to projected water, simulated policy count, exposure gap score (indetify places where exposure rises faster than insurance).

3. Research

I looked at serveral studies and research that discussed how insurance companies are using spatial data and risk models to giude their management. For example, CARTO created a map where it showed how insurers combine assest locations with flood zones, weather info, future flooding to identify exposure and make decsions. Adding to thing, NYT investgated the US homeowners insurance market and reported that climate related disaster have contributed to insurer losses, premium increases and rediced coverage or withdrwal. Making insurance unprofitable in some areas. In Orleans Parish, Louisiana, approximately 1 in 26 policies were nonrenewed in 2023, a 766.5% increase from 2018.



Sources

- Puente, Estefania. "Grow Your Insurance Data Analytics with 12 Interactive Maps." CARTO, 2024.
- The New York Times. "The Home Insurance Crunch: See What's Happening in Your State." 2024.
- The New York Times. "What's Going On in This Graph? Homeowners Insurance." 2026.

"What's Going On in This Graph? | Homeowners Insurance." The New York Times, 22 Jan. 2026, <https://www.nytimes.com/2026/01/22/learning/whats-going-on-in-this-graph-feb-4-2026.html>

4. My Record

The simulation begins in 2025. One row represents one census tract, ZIP code, or parish in the New Orleans region or coastal Louisiana. The starting record combines the number of active NFIP policies in the area with nearby historic land-loss conditions.

5. My Rule

The simulation would connect changing coastline to a simulated change in insurance availability.

Rule 1> Calculate policy trend and the baseline policy number using the 2020 FEMA records. This would ask what would happen if each geography continued changing at the same rate it showed between 2020 and 2025, without any new adjustment for projected coastal land loss?

$$\text{Policies}_{i,t+5} = \text{Policies}_{i,t} \times (1 + \text{observed local 2020–2025 policy-change rate})$$

Rule 2> Project Future Land Loss The historical land loss data comes from USGS

$$\text{Projected land area}_{j,t+5} = \text{Land area}_{j,t} - (\text{Historic annual land-loss rate}_j \times 5)$$

From this result the distance to a newly projected water, the area/percentage lost and wehter the geography is protected or not can be assigned.

Rule 3> Adjusting for coastal exposure B is observed policy change, W is new water penalty, L is land loss, P is protection credit

$$\text{Policies}_{t+5} = \text{Policies}_t \times (1 - B - W - L + P)$$

The values I am least sure of are the 1-kilometer distance threshold, the –2-point water adjustment, the –2/-5-point land-loss adjustments, and the +1-point protection adjustment. These values are not forecasts; they are deliberately visible assumptions for testing possible outcomes.

Geography	Year	Starting policies	Observed 2020–2025 trend	New water within 1 km?	Land-loss category	Protected?
LA-TRACT-01	2025	185	–5%	Yes	High	No
LA-TRACT-01	2030	163	–5%	Yes	High	No
LA-TRACT-01	2035	143	–5%	Yes	Moderate	No

Calculation	Policies at next step
$185 \times [1 - .05 - .02 - .05 + 0]$	163
$163 \times [1 - .05 - .02 - .05 + 0]$	143
$143 \times [1 - .05 - .02 - .02 + 0]$	133

5. My Run

The model moves in five year increments. At each five year step the land loss rule would apply, so it would generate projected land loss. The first model will be using fixed rules, the next times a different scenario would be created changing values like severe retreat, stable coverage or medium retreat.

6. What the data would generate

The model would generate a new number for each zip code every five years. So there would be a new row with the updated distance to projected water, about of projected land loss, if there is a restoration condtion and the number of active NFIP policies. Using this new row, the output would indentify places where policy counts remain stabel or places where policies decline as exposure increased and gaps. Visually this could be represented in maps every 10 years.

7. What the data cant see

- The model cannot predict real NFIP policy cancellations, renewals, premiums, coverage amounts, or insurer decisions.
- FEMA's NFIP, private flood insurance have different rules
- A simulated drop in policies could reflect migration, property loss, abandonment, unaffordability, private insurance substitution, nonrenewal, or changes in data reporting. Model cant distinguish what the cause is.
- Policy count does not show uninsured households, renters, businesses, people who cannot afford insurance
- The model does not represent income, race, age, disability, tenure, mortgage status, property value, or other factors shaping insurance access and vulnerability.
- It does not calculate flood depth, storm surge, rain-driven flooding, hurricane damage, levee performance, drainage, or property-level risk.
- It simplifies future land loss by extending historic patterns

